TEXORA · SESSIONAL NOTES · STUDY SHEET

Apparel Production: Basic Shirt, Care Labelling and Industrial Visit

The term's work assembled: producing a complete basic shirt on industrial machines through its operation sequence, reading and drawing the five-symbol international care labelling code that the finished shirt must carry, and the industrial visit that closes the study.

TE-2208 · APPAREL MANUFACTURING TECHNOLOGY-1 SESSIONAL · STUDY 10 · LEVEL 2 TERM 2

1. Aim and place of the study

Aim: to produce a basic shirt from the cut components of the earlier studies by following its operation sequence (breakdown), to apply the international care labelling code correctly to the finished garment, and to record the industrial visit made as part of this study. Everything studied separately — patterns (Studies 3–5), plain and stitch-forming machines (Studies 6–8), cutting (Study 9) — is used here in one garment.

2. Operation sequence of the basic shirt

The sequence is the order in which operations must occur: parts are prepared first, then assembled from the inside of the garment outward — yoke and shoulders before collar, sleeves set before side seams close the tube, closures and finishing last. Reversing two dependent operations either makes assembly impossible or buries an unfinished edge.

NO.	OPERATION	MACHINE / EQUIPMENT
1	Crease and prepare the front plackets; mark pocket position	Pressing iron; SNLS if the placket is stitched
2	Hem the pocket top and attach the pocket to the left front	Single-needle lockstitch (SNLS)
3	Make the collar: join fall and band parts, trim, turn, press and topstitch	SNLS; turning and pressing aids
4	Make the cuffs (with sleeve plackets prepared)	SNLS

5	Attach the yoke to the back, enclosing the back top edge (double-yoke method)	SNLS
6	Join the shoulders — fronts to yoke/back	SNLS (seam safety-stitched or overlocked by spec)
7	Attach the collar to the neckline and close the band ends	SNLS
8	Set the sleeves into the armholes, distributing the cap ease	SNLS (in industry often chain stitch, feed-off-the-arm)
9	Close the side seams and sleeve underarm seams in one pass	Chain stitch or SNLS, per specification
10	Attach the cuffs to the sleeve ends, closing the placket neatly	SNLS
11	Hem the bottom of the shirt	SNLS with hem guide
12	Make buttonholes (placket, band, cuffs)	Buttonhole machine
13	Attach buttons in line with the buttonholes	Button attaching machine
14	Trim threads, remove marks, final check and measure against the spec	Manual; measuring table (Study 2 method)
15	Press the shirt, fold, tag and pack	Steam iron / pressing table

At each operation the work is checked before it passes on: a fault sewn over by the next operation is found at operation 14 at the earliest, with the whole garment to unpick. This is the production form of the rule running through the sessional — verify at the stage where correction is still cheap.

3. The international care labelling code

A finished garment must tell its user how to care for it, in symbols that cross all languages. The international code (ISO 3758, administered in its registered form by GINETEX) uses **five basic symbols, always in the same order: washing → bleaching → drying → ironing → professional care**. Modifications are drawn inside or beside the symbol: numbers give temperatures, dots give heat levels, bars under a symbol call for gentler action, and a cross (St Andrew's cross) over any symbol prohibits that treatment.

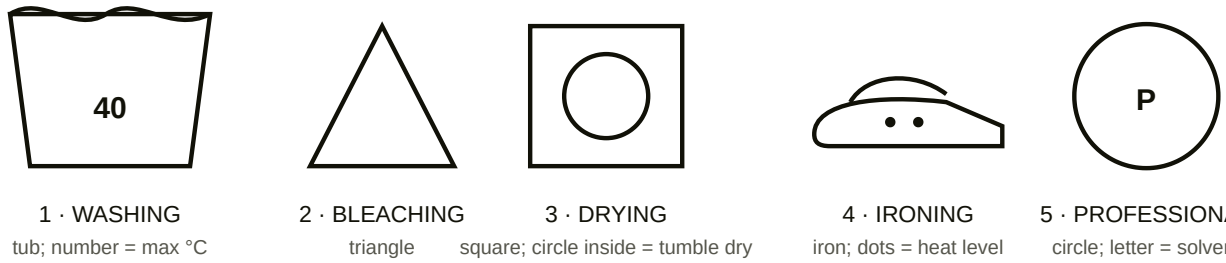


Figure 1 — The five basic care symbols in their fixed order, in a common form for a washable cotton shirt: wash at 40 °C, bleaching allowed, tumble dry, iron at medium heat (two dots), professional dry cleaning with solvent P.

SYMBOL	BASIC MEANING	COMMON MODIFICATIONS
Washing (tub)	Domestic washing in water. The number inside is the maximum wash temperature in °C (e.g. 40).	A hand drawn in the tub = hand wash only (max 40 °C). One bar under the tub = mild process; two bars = very mild. Crossed tub = do not wash.
Bleaching (triangle)	An empty triangle: bleaching allowed.	Two diagonal lines inside = oxygen (non-chlorine) bleach only. Crossed triangle = do not bleach.
Drying (square)	A circle inside the square: tumble drying allowed; dots in the circle give the heat (one dot low, two dots normal).	Without the circle, lines inside the square prescribe natural drying: one vertical line = line dry, two = drip line dry; one horizontal line = dry flat, two = drip dry flat. Crossed = do not tumble dry (over the tumble symbol).
Ironing (iron)	Ironing allowed. Dots set the soleplate temperature: one dot 110 °C, two dots 150 °C, three dots 200 °C.	Crossed iron = do not iron. A crossed steam burst added = no steam.
Professional care (circle)	Professional dry or wet cleaning. The letter inside names the process: P = dry clean with the usual solvents (perchloroethylene and hydrocarbons), F = the gentler hydrocarbon-only process, W = professional wet cleaning.	A bar under the circle = gentle process; two bars = very gentle. Crossed circle = do not dry clean.

Worked reading. The label in Figure 1 instructs: wash in water at up to 40 °C; bleach when needed; tumble dry; iron at medium heat (150 °C, two dots); and the garment may be professionally dry cleaned

with P solvents. The five symbols are read left to right in their fixed order — a label that rearranges them is wrongly printed, however correct each symbol is.

4. The industrial visit

The study closes with a visit to at least one working textile or apparel industry, as the syllabus requires for this course group. The visit is a study exercise, not a tour: the student carries the laboratory's questions into the factory.

- **Observe the layout** against Study 1: where do the store, cutting, sewing and finishing sit, and does material flow in one direction?
- **Observe the machines** against Studies 6–9: which stitch types are in use on which seams, which cutting machines divide the cutting work, and how lines are balanced.
- **Observe quality and records:** in-line and end-line inspection points, spec sheets at the line, bundle tickets, and the care labels being attached — the five symbols of §3, in order, on a real label.
- **Write the report** on return: factory profile, production flow observed section by section, machines and stitches identified, and a short comparison with the teaching laboratory — what the factory does that the laboratory should copy, and what it does differently because of scale.

The report is the record of this study's visit and forms part of the sessional record for the course.

5. Precautions and good practice

- Follow the operation sequence; never close a seam that a later operation must open to work inside the garment.
- Check each operation before passing the work on — assembly hides faults exactly where they are most expensive to reach.
- Print care symbols only from the standard set, in the standard order, and only the treatments the garment can actually survive; an over-promising label damages garments and returns claims against the factory.
- On the visit: follow the host factory's safety rules, stay with the guide, and never touch machines or work in progress.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. State the five care symbols in their fixed order and the process each governs.

2. Read this label aloud: tub with "30", triangle with two diagonal lines, square with a crossed circle inside, iron with one dot, crossed circle. What care do you give the garment?
3. What do the dots inside the iron symbol mean, and what temperatures do one, two and three dots set?
4. In the shirt's operation sequence, why is the collar attached (operation 7) before the sleeves are set (operation 8), and the buttonholes made (12) after assembly?
5. Which operations of the sequence consume the products of Studies 7 and 8, and which stitch types do they use?
6. What four observations must the industrial-visit report record?
7. A label shows the iron symbol first and the tub second, all symbols individually correct. Is the label acceptable? Why?

Answer key

1. Washing (tub), bleaching (triangle), drying (square), ironing (iron), professional care (circle).
2. Wash at up to 30 °C; oxygen (non-chlorine) bleach only; do not tumble dry; iron at low heat, 110 °C; do not dry clean.
3. They set the maximum soleplate temperature: one dot 110 °C, two dots 150 °C, three dots 200 °C.
4. The collar closes the neckline, which must be finished while the garment is still open and flat enough to handle; once the sleeves are set and the sides closed the shirt is a tube and the neck is hard to work on. Buttonholes are positioned on the finished, assembled fronts and cuffs so that holes and (later) buttons align on the garment as worn — making them earlier would fix positions before the parts are finally joined.
5. The seaming operations (2, 3, 5–11) use the lock stitch 301 of Study 7 on the SNLS machine, and where the specification allows, the side-seam and armhole closing uses the chain stitch 401 of Study 8 for its elasticity and uninterrupted long seams.
6. The factory layout and material flow; the machines and stitch types in use on each seam; the quality and record system (inspection points, spec sheets, bundle tickets, labels); and a comparison with the teaching laboratory with lessons to copy.
7. No. The code fixes the order — washing, bleaching, drying, ironing, professional care — so that any reader anywhere reads any label the same way; a reordered label breaks the standard even if every symbol is drawn correctly.
